

From uniform control to one-sided ES certificates

An original divisor state succeeds precisely when its odd return denominator D is one. Keep the discrete spectrum, not merely the interval containing it:

$$D^{-1} \in \{1, 1/3, 1/5, 1/7, \dots\}.$$

An explicit signed filter

$$\Phi_j(x) = x \prod_{\nu=1}^j \frac{(2\nu+1)x-1}{2\nu}, \quad \Phi_{2r+1} \leq \mathbf{1}_{\{1\}} \leq \Phi_{2r}.$$

$$\|\Phi_j - \mathbf{1}_{\{1\}}\|_{\Sigma} \leq \frac{1}{2^j j! (2j+3)}, \quad \sum_k |[x^k] \Phi_j| = j+1.$$

Factorial error decay; only linear amplification of absolute moment errors. With the original arithmetic row mass retained, this gives certified lower and upper counts for every canonical row defined here, for primes $p \equiv 1 \pmod{4}$.

A genuine hard-prime test

$$p = 19\,748\,674\,681 \equiv 1 \pmod{840}, \quad R = 135.$$

Its complete shell has 2,106 original states and exactly one hit. The optimal aggregate first-degree signed test is negative:

$$L_1 = -\frac{50887012}{144801015} < 0.$$

Retaining the discrete spectrum repairs the test:

$$\Phi_3(x) = \frac{105x^4 - 71x^3 + 15x^2 - x}{48},$$

$$\sum_{\alpha} \Phi_3(D_{\alpha}^{-1}) = \frac{1297691}{2460375} > 0.$$

The corresponding original exterior marking is

$$(h, r, s, \kappa) = (26, 2336, 81289, 341725215823).$$

Complete proofs, deterministic prime certificates, exact source weights, ordered denominator inverses, JSON and a standard-library verifier accompany the two three-page preprints. This is not a proof of universal ES, a least counterexample to the moment shortcut, or a claim of new prime coverage.